

Experiment 3

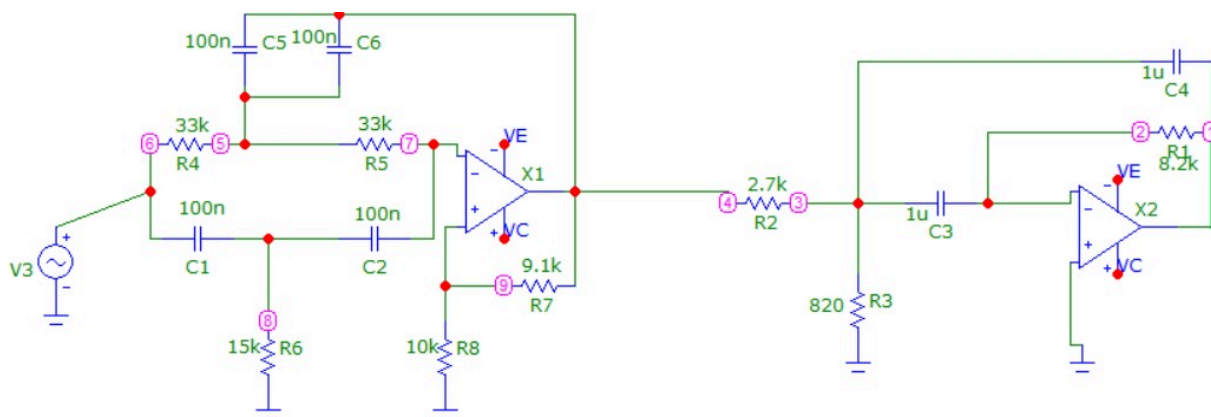
Measurement of ECG signals

We had to design a twin-T notch filter and a band pass filter optimised to measure ECG signals. This had to be followed by routing the circuit on a PCB. After the fabrication of the PCB, we implemented the circuit on the PCB using an instrumentation amplifier and the components available in the lab.

Group Members

1. Akshat Dhote (230095)
2. Sattyadeep Garg (230935)
3. Shivam Yadav (230972)
4. Shivram Khatik (230979)

Circuit Design and Simulation



CIRCUIT DESIGNED

Note: The notch-filter used in the circuit was highly sensitive to precise values of resistors. Due to the availability of components in the lab, the circuit notch appears at 77 Hz instead of the intended 50 Hz.

Notch Filter Design

To achieve a notch at 50Hz, we use the fundamental frequency formula:

$$f_c = 1/2\pi RC$$

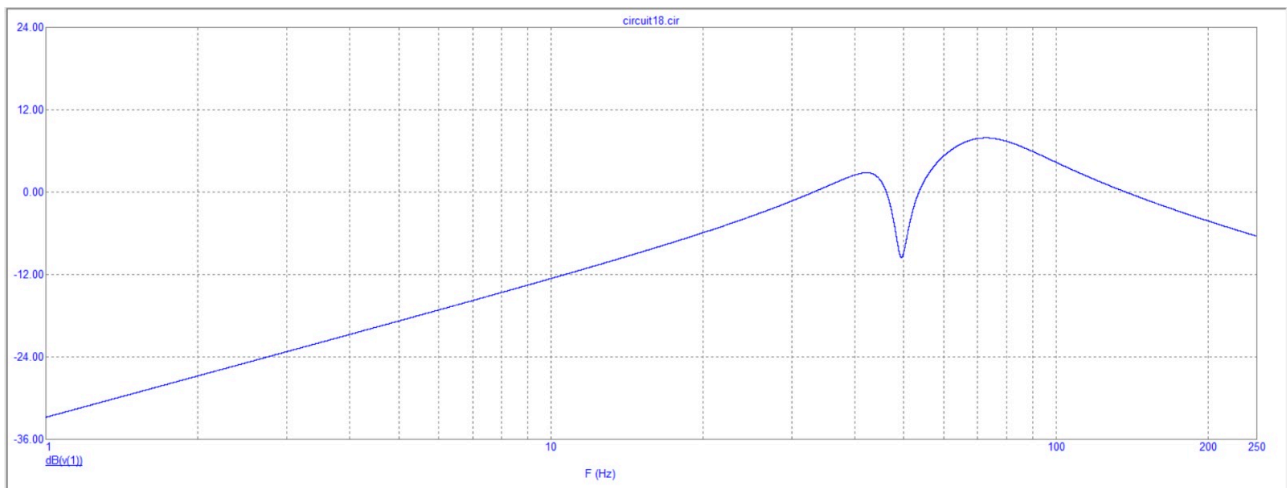
Let's say we chose $C=100$ nF.

Calculate R:

$$R = 1 / (2\pi \cdot 50 \text{ Hz} \cdot 100 \text{ nF}) \approx 31.8 \text{ k}\Omega$$

For the notch to be efficient, the T-branches must be symmetrical:

- $R4 = R5 = R$
- $R6 = R/2 \approx 15 \text{ k}\Omega$
- $C1 = C2 = C$
- $C5 + C6 = 2C = 200 \text{ nF}$



SIMULATED CIRCUIT RESPONSE

Lab Work

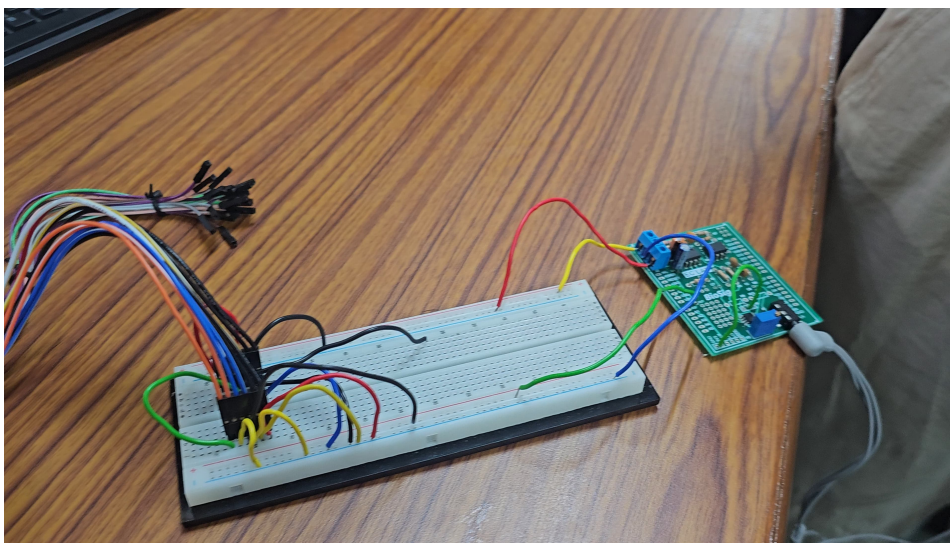
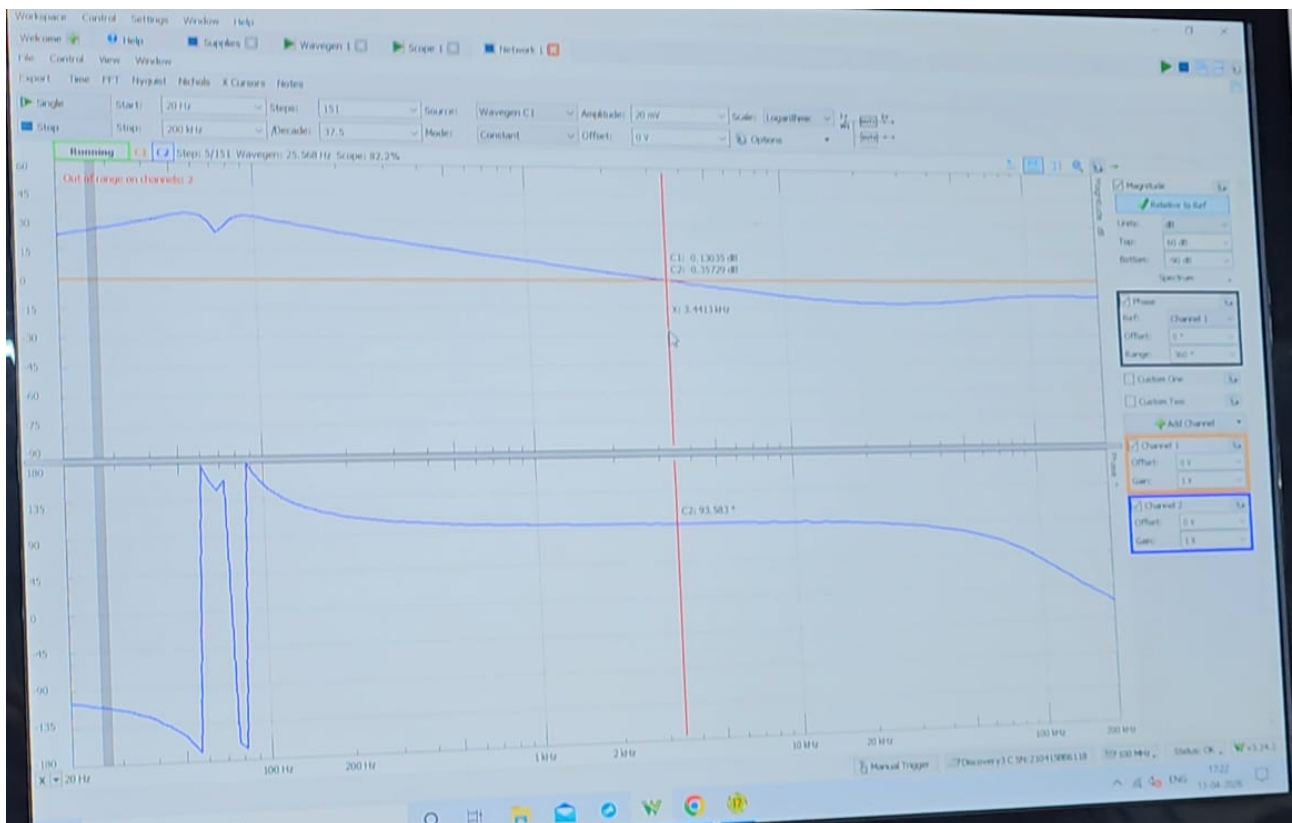


IMAGE SHOWING THE FINAL PCB AND CIRCUIT ASSEMBLY



IMAGE SHOWING THE REAL TIME MEASUREMENT OF ECG SIGNAL.



FREQUENCY RESPONSE OF THE FINAL CIRCUIT, WITH NOTCH AT 50 HZ